

Bike Station Utilization Analysis

Descriptive Analysis

- Provides an overview of bike occupancy, station status, and bike availability.
- Some stations show significantly higher bike occupancy and availability.
- Bike availability varies across different stations.
- The dashboard highlights top-performing and low-performing stations.
- Bike-sharing usage increased significantly in 2025.
- Available bikes reached **8.7K**, while low availability reached **10.8K**.
- High usage stations account for **53.33%** of total stations.
- Medium stands have the highest bike availability with **12.7K** bikes.

Diagnostic Analysis

- High occupancy stations may be located in high-demand areas.
- Low bike availability may indicate increased customer usage.
- Stations with excess available bikes may experience lower demand.
- Closed stations may be affected by maintenance or operational issues.
- The analysis identifies imbalance in bike distribution across stations.
- Increasing demand is causing higher pressure on stations.
- High-utilization stations face frequent low bike availability.
- Medium stands perform more efficiently compared to small and large stands.

Predictive Analysis

- High-demand stations are likely to remain busy in the future.
- Some stations may experience bike shortages during peak hours.
- Stations with consistently high bike availability may become underutilized.
- Operational performance is expected to remain stable if station availability is maintained.
- Bike demand and station utilization are expected to grow further.
- High-demand stations may experience more shortages in the future.
- Medium stands will likely remain the preferred expansion model.

Prescriptive Analysis

- Allocate additional bikes to high-demand stations.
- Implement bike redistribution strategies for balanced availability.
- Use preventive maintenance to reduce station closures.
- Enable real-time monitoring and alert systems.
- Apply demand-based resource allocation to improve efficiency.
- Improve real-time bike redistribution in busy stations.
- Increase capacity in high-usage areas.
- Expand medium-sized stands for better operational efficiency.
- Use demand forecasting for better resource planning.
